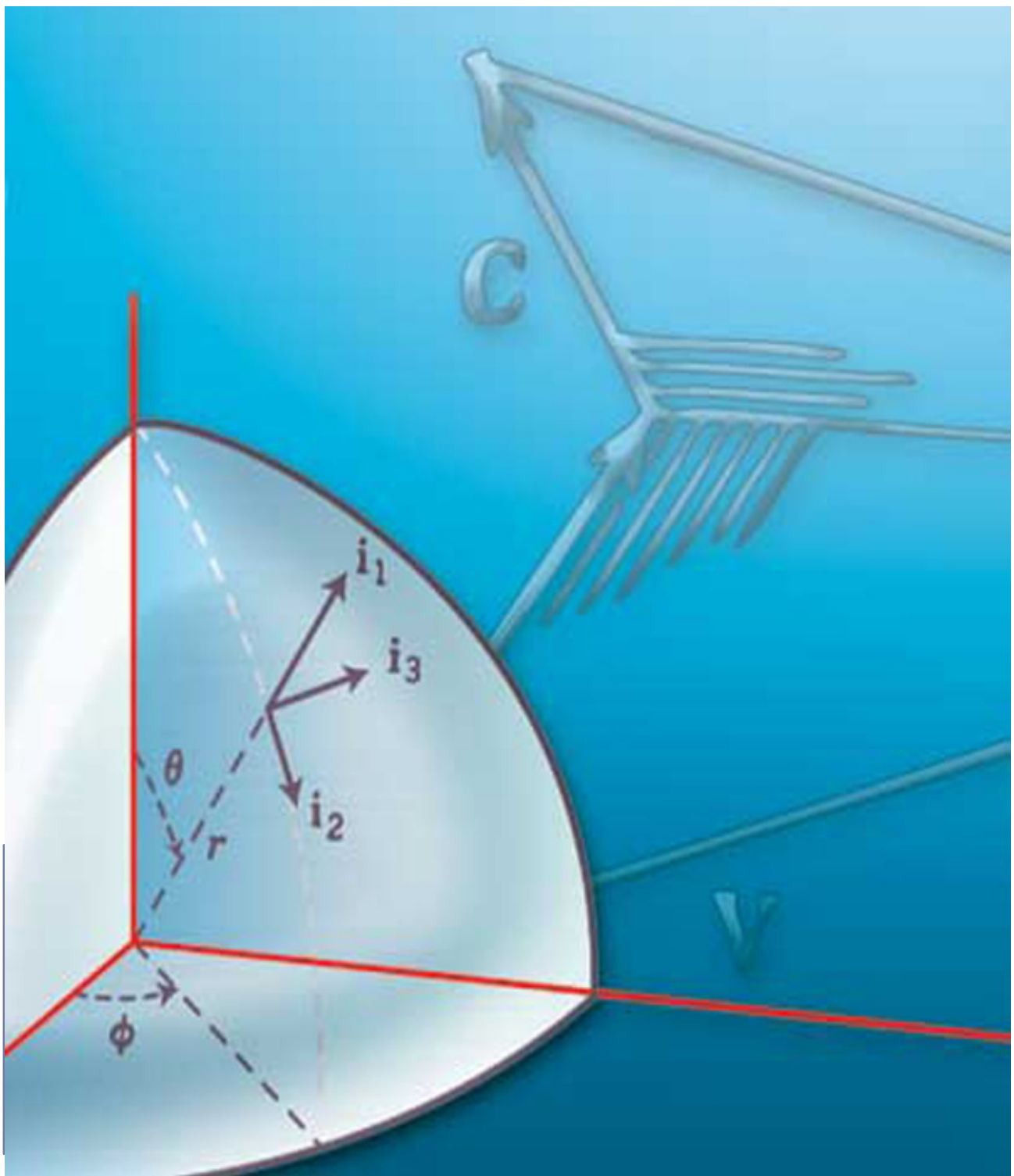




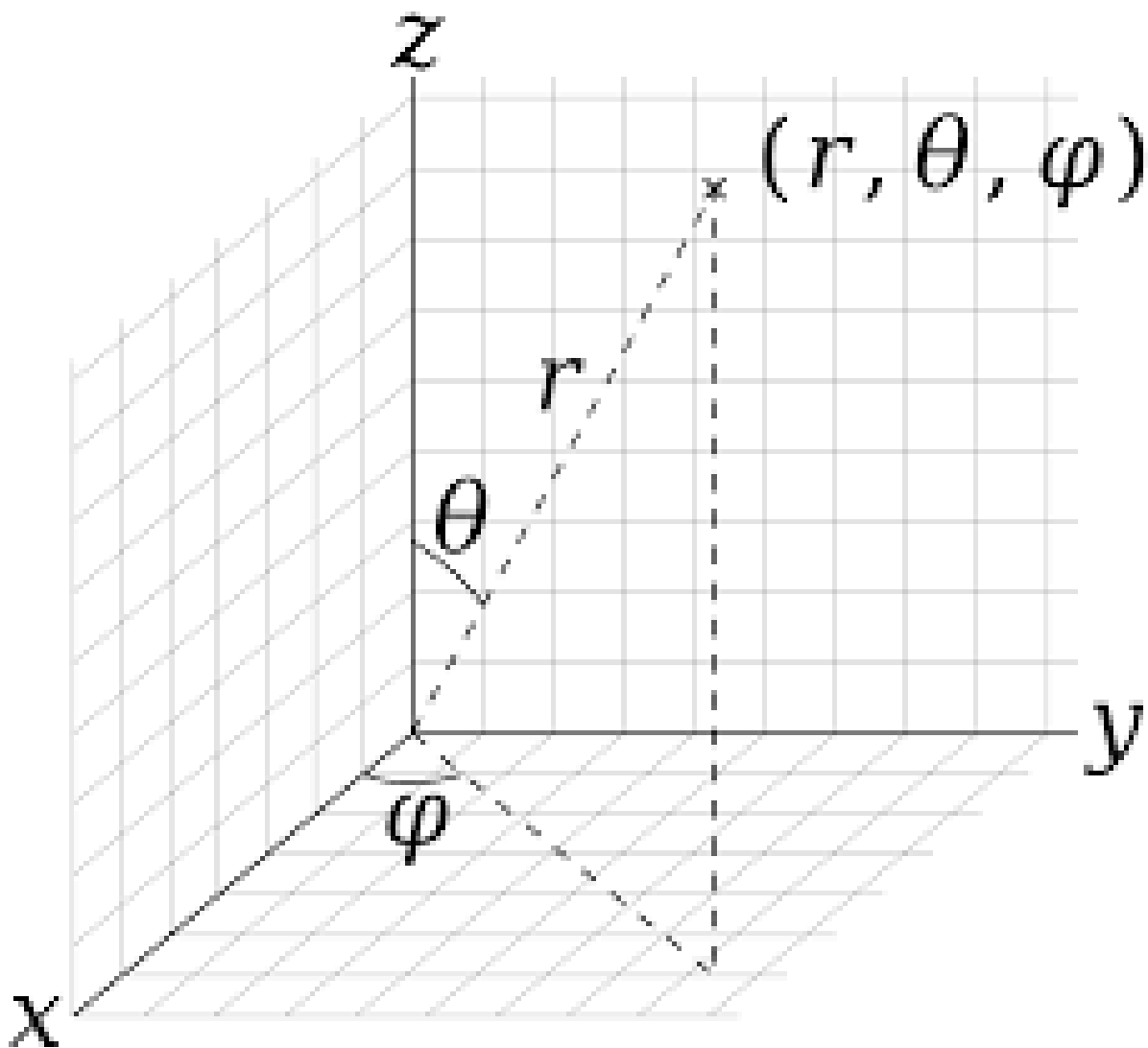
VECTOR ANALYSIS



Edit with WPS Office

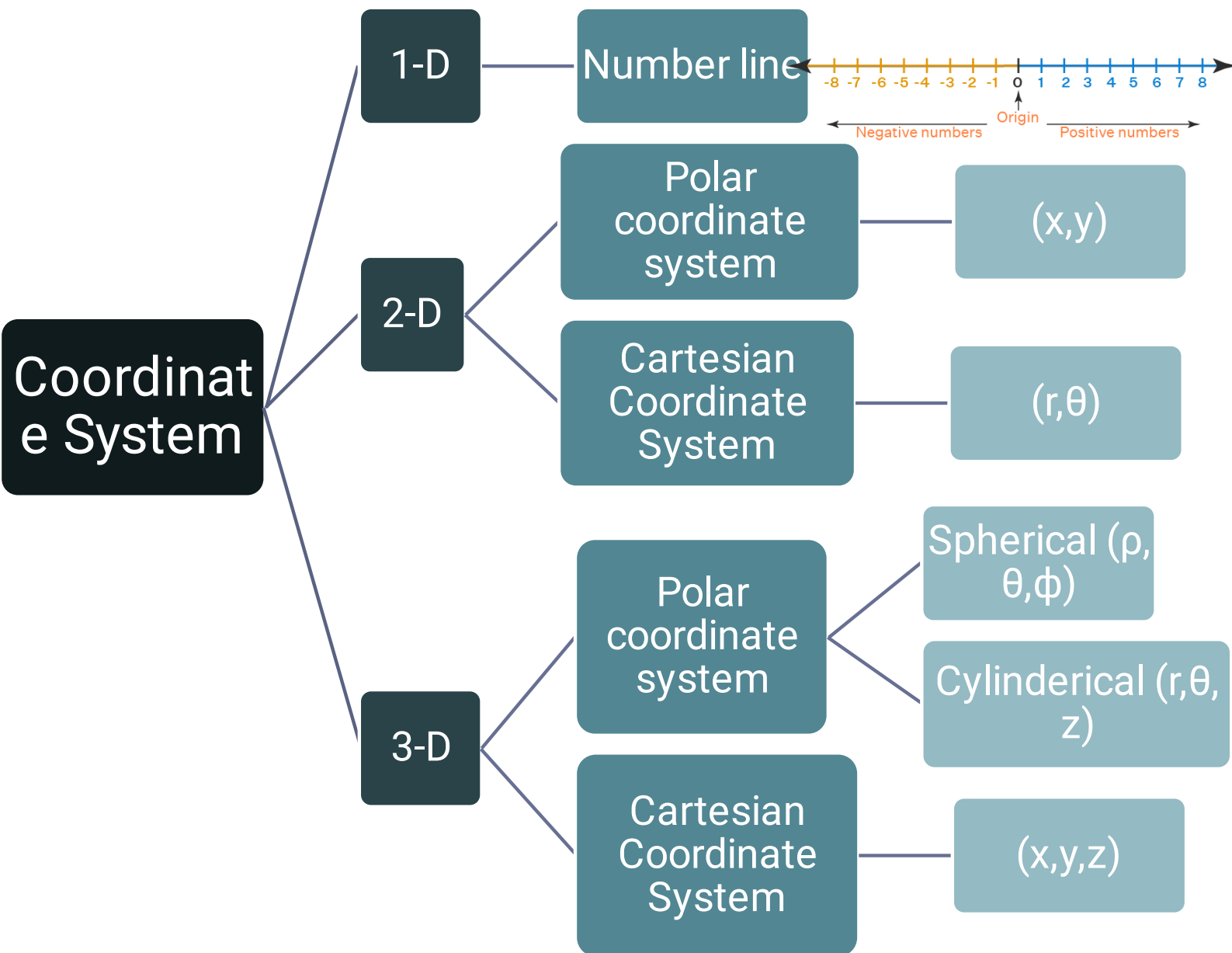
COORDINATE SYSTEM:

Used to identify the location of objects or points in space.



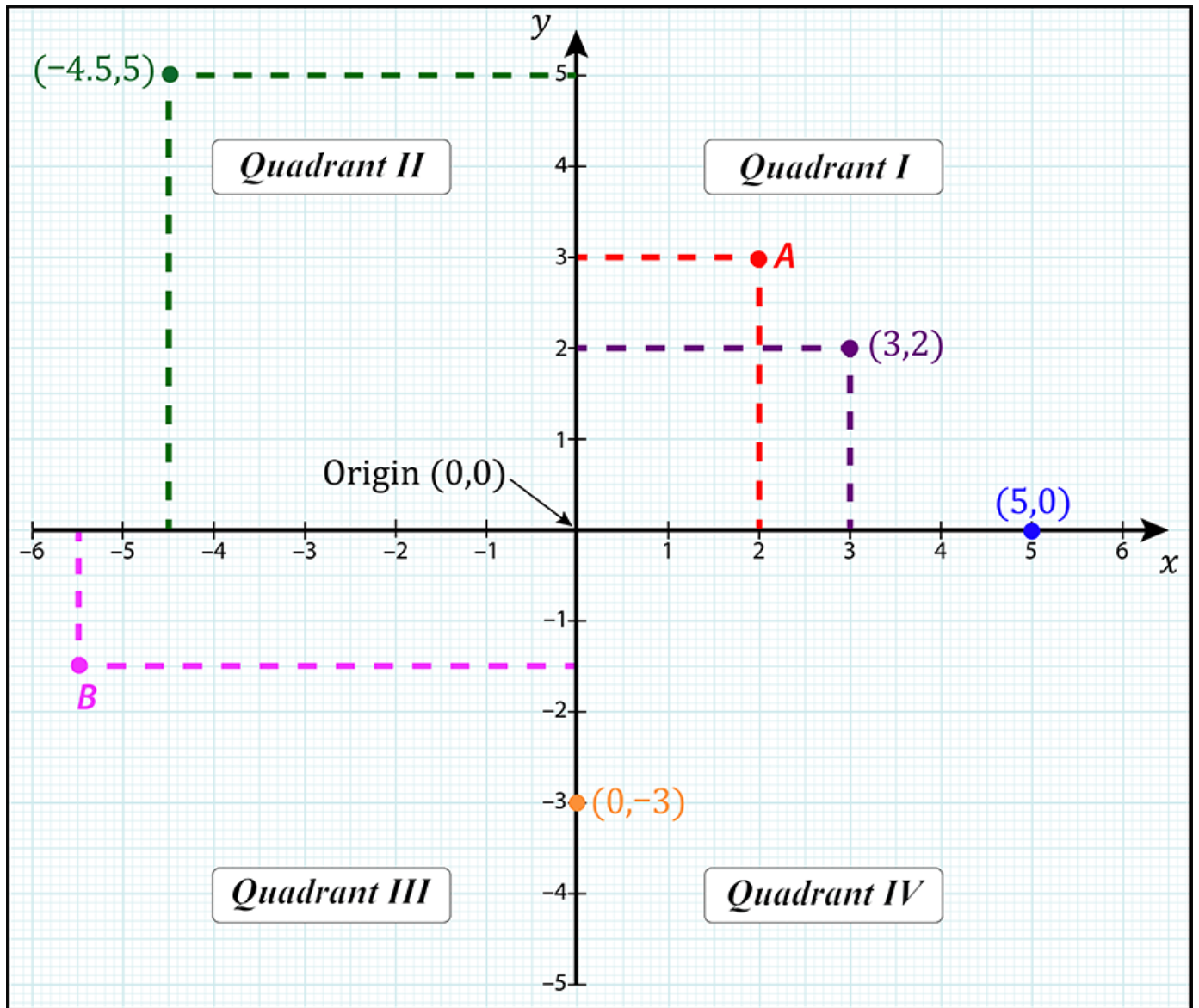
Pictorial Representation





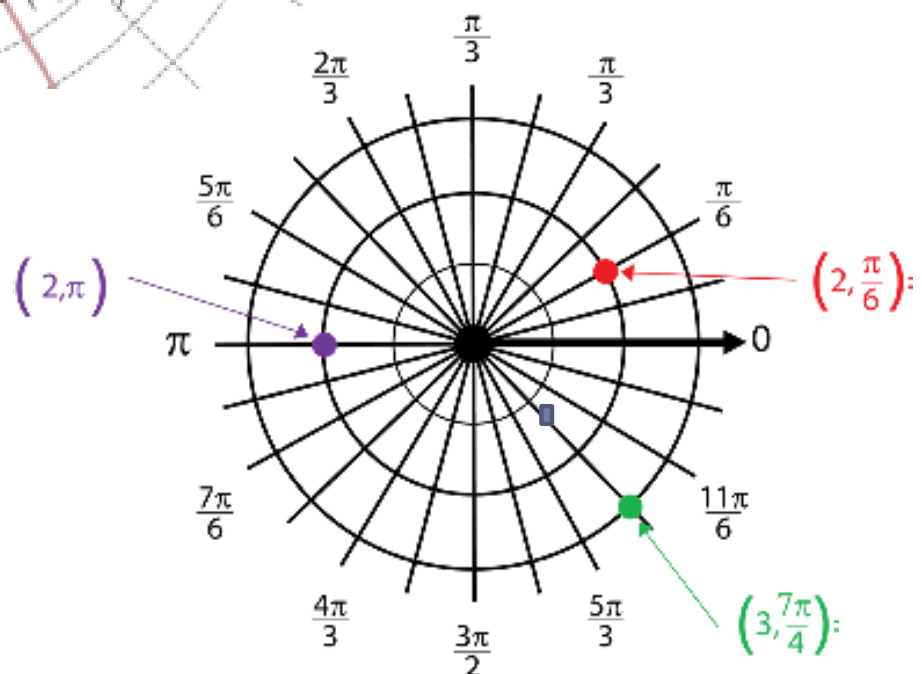
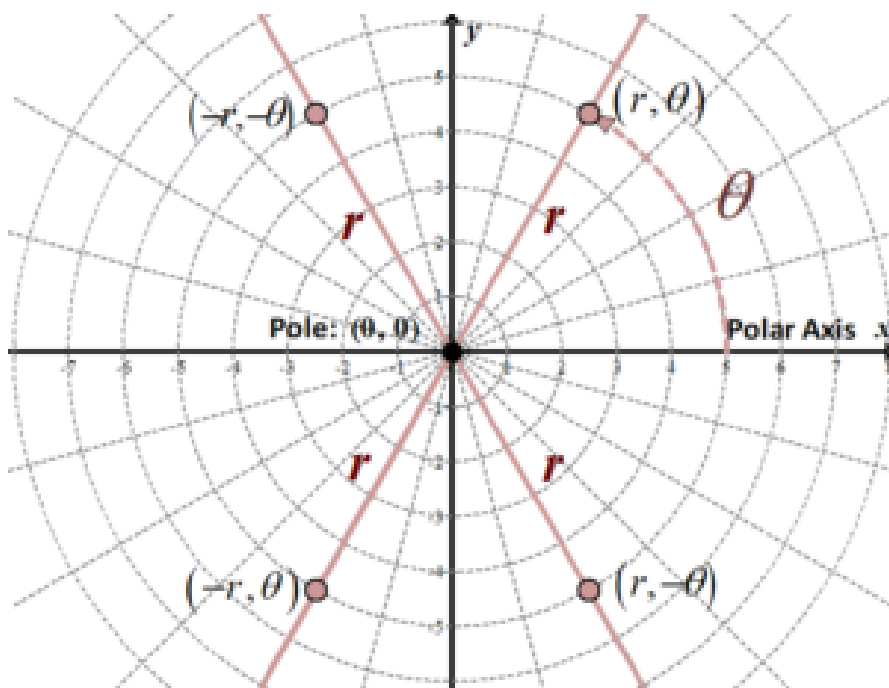
1. Cartesian Coordinate System:

- ▣ Locate object with respect to x,y coordinates.



2. Polar Coordinate System:

- Location of object is determined by the distance (r) and the angle (θ) from the reference point (i.e. origin = 0)



Conversions:

► Cylindrical → Cartesian:

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

► Cartesian → Cylindrical:

$$r^2 = x^2 + y^2$$

$$\tan \theta = \frac{y}{x}$$

$$z = z$$

► Spherical → Cartesian:

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

► Cartesian → Spherical:

$$\rho^2 = x^2 + y^2 + z^2$$

$$\tan \theta = \frac{y}{x}$$

$$\phi = \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right)$$



PROBLEM:

1. Convert the point $(2, \frac{4\pi}{3}, 8)$ from cylindrical system to Cartesian system.
2. Convert the point $(\sqrt{3}, 1, 4)$ from Cartesian to :
 1. Spherical System
 2. Cylindrical system



VECTORS

Vector Algebra

(+, -, dot product, Cross product)

Vector Calculus

Differentiation

Ordinary

Partial

Curl

Gradient

Divergence

Integration

Line

Surface

Volume



Scalar & Vector Quantities:

- ⊠ The quantities that can completely be explained by magnitude only are called scalar quantities. (mass, time, speed, length, etc)
- ⊠ The quantities that are explained by magnitude and direction are called **vector** quantities. (Velocity, Acceleration, torque, displacement etc)



Types of Vectors:

1. Position vector ($\vec{r}$)
2. Free vector
3. Null vector
4. Magnitude of vector $\vec{V} = \sqrt{x^2 + y^2 + z^2}$
5. Resultant vector
6. Unit vector $\hat{A} = \frac{\vec{A}}{|\vec{A}|}$



Ordinary Differentiation:-

- ☒ Deals with functions of a single variable (e.g., $f(x)$)
- ☒ Finds the rate of change of the function with respect to that variable (e.g., $f'(x)$ or df/dx)
- ☒ Geometrically, represents the slope of the tangent line to the function at a point



Partial Differentiation:-

- ☒ Deals with functions of multiple variables (e.g., $f(x, y, z)$)
- ☒ Finds the rate of change of the function with respect to one variable, while keeping the other variables constant (e.g., $\partial f / \partial x$, $\partial f / \partial y$, etc.)
- ☒ Geometrically, represents the slope of the tangent line to the function in the direction of the variable being differentiated, while keeping the other variables fixed.
- ☒ **Key differences:-**
- ☒ OD involves a single variable, while PD involves multiple variables. OD finds the total rate of change, while PD finds the rate of change with respect to a specific variable.



Ordinary Vector Differential



$$\blacktriangleright \frac{d(A+B)}{dt} = \frac{dA}{dt} + \frac{dB}{dt}$$

$$\blacktriangleright \frac{d(A \cdot B)}{dt} = A \cdot \frac{dB}{dt} + B \cdot \frac{dA}{dt}$$

$$\blacktriangleright \frac{d(A \times B)}{dt} = A \times \frac{dB}{dt} + B \times \frac{dA}{dt}$$

$$\blacktriangleright \frac{d(\emptyset A)}{dt} = \emptyset \cdot \frac{dA}{dt}, \text{ (where: } \emptyset = \text{ scalar quantity)}$$



Example 12. A particle moves along the curve $\vec{r} = (t^3 - 4t)\hat{i} + (t^2 + 4t)\hat{j} + (8t^2 - 3t^3)\hat{k}$, where t is the time. Find the magnitude of the tangential components of its acceleration at $t = 2$.

(Nagpur University, Summer 2005)

Solution. We have, $\vec{r} = (t^3 - 4t)\hat{i} + (t^2 + 4t)\hat{j} + (8t^2 - 3t^3)\hat{k}$

$$\text{Velocity} = \frac{d\vec{r}}{dt} = (3t^2 - 4)\hat{i} + (2t + 4)\hat{j} + (16t - 9t^2)\hat{k}$$

At $t = 2$, Velocity = $8\hat{i} + 8\hat{j} - 4\hat{k}$

$$\text{Acceleration} = \vec{a} = \frac{d^2\vec{r}}{dt^2} = 6t\hat{i} + 2\hat{j} + (16 - 18t)\hat{k}$$

At $t = 2$ $\vec{a} = 12\hat{i} + 2\hat{j} - 20\hat{k}$

The direction of velocity is along tangent.

So the tangent vector is velocity.

$$\text{Unit tangent vector, } \hat{T} = \frac{\vec{v}}{|\vec{v}|} = \frac{8\hat{i} + 8\hat{j} - 4\hat{k}}{\sqrt{64 + 64 + 16}} = \frac{8\hat{i} + 8\hat{j} - 4\hat{k}}{12} = \frac{2\hat{i} + 2\hat{j} - \hat{k}}{3}$$

Tangential component of acceleration, $a_t = \vec{a} \cdot \hat{T}$

$$= (12\hat{i} + 2\hat{j} - 20\hat{k}) \cdot \frac{2\hat{i} + 2\hat{j} - \hat{k}}{3} = \frac{24 + 4 + 20}{3} = \frac{48}{3} = 16 \text{ Ans.}$$

Practice Problems:

- The coordinates of a moving particle are given by $x = 4t - \frac{t^2}{2}$ and $y = 3 + 6t - \frac{t^3}{6}$. Find the velocity and acceleration of the particle when $t = 2$ secs. **Ans.** 4.47, 2.24
- A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$ and $z = 3t - 5$ where t is the time. Find the components of its velocity and acceleration at time $t = 1$, in the direction $\hat{i} - 3\hat{j} + 2\hat{k}$. (Nagpur, Summer 2001) **Ans.** $\frac{8\sqrt{14}}{7}, -\frac{\sqrt{14}}{7}$
- Find the unit tangent and unit normal vector at $t = 2$ on the curve $x = t^2 - 1$, $y = 4t - 3$, $z = 2t^2 - 6t$ where t is any variable. **Ans.** $\frac{1}{3}(2\hat{i} + 2\hat{j} + \hat{k}), \frac{1}{3\sqrt{5}}(2\hat{i} + 2\hat{k})$

Problems:

- ▶ If $\vec{P} = 2i - k$, $\vec{Q} = 5j + 2k$. Find the following:
 - ▶ $|P|$, $|Q|$, $P + Q$, $P - Q$, $|P - Q|$

- ▶ Given that, $\vec{A} = i + 2j + 3k$ and $\vec{B} = 2i + 4j - k$. Find:
 - ▶ $A \cdot B$, $A \times B$, The angle between A & B .

- ▶ If $\vec{A} = i + j + k$ and $\vec{B} = 2i - j + 3k$. Find a unit vector parallel $|A - 2B|$



Partial Differential:

Vector Differential Operator Del i.e. ∇

The vector differential operator Del is denoted by ∇ . It is defined as

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

⊠ Gradient: Maximum rate of increase.

If $\phi(x, y, z)$ be a scalar function then $\hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$ is called the gradient of the scalar function ϕ .

And is denoted by $\text{grad } \phi$.

Thus,

$$\text{grad } \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$\text{grad } \phi = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \phi(x, y, z)$$

$$\text{grad } \phi = \nabla \phi \quad (\nabla \text{ is read del or nebla})$$



Gradient Problems:

- ▣ If $\phi(x, y, z) = 3x^2 - y^3z^3$. Find Gradient at point $(1, -2, -1)$.



Directional Derivative of Gradient:

☒ $\nabla \phi \cdot \hat{a}$

- Find the directional derivative in direction $2\hat{i} + 3\hat{j} - 2\hat{k}$. If $\phi(x, y, z) = 2xy + 2yz$ at point $(0, 1, 2)$.

Practice Problem:

The temperature of the points in space is given by $T(x, y, z) = x^2 + y^2 - z$. A mosquito located at $(1, 1, 2)$ desires to fly in such a direction that it will get warm as soon as possible. In what direction should it move?

Ans. $\frac{1}{3}(2\hat{i} + 2\hat{j} - \hat{k})$



⊠ Divergence: How much the quantity

$$\text{div } \mathbf{v} = \nabla \cdot \mathbf{v} = \left[i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right] \cdot [iv_1 + jv_2 + kv_3]$$

$$\nabla \cdot \mathbf{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

Results Interpretation:

- ⊠ +ve = Source field
- ⊠ -ve = Sink field
- ⊠ 0 = Solenoid field

⊠ Curl: Rotation of a vector.

Results Interpretation:

- ⊠ 0 = Irrotational field

$$\nabla \times \underline{\underline{A}} = \begin{vmatrix} \underline{\underline{a}}_x & \underline{\underline{a}}_y & \underline{\underline{a}}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$= \underline{\underline{a}}_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \underline{\underline{a}}_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \underline{\underline{a}}_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$